

YEAR 11 GENERAL HUMAN BIOLOGY

GEHBY Task 9: DNA, Mitosis and Meiosis

NAME: Marking Key

TEACHER: _____

DATE: _____

SCORES:

MC: _____ /15

SA: _____ /15

TOTAL: _____ /30

_____ %

Section A: Multiple choice (15 Marks)

Answer all questions by circling the most correct answer on the multiple choice answer sheet.

1. ☒ a b c d
2. a b ☒ c d
3. a ☒ b c d
4. a b c ☒ d
5. a b ☒ c d
6. a b ☒ c d
7. a ☒ b c d
8. ☒ a b c d
9. a b c ☒ d
10. a b c ☒ d

Section B Short Answer**25 Marks****Question 1**

How many chromosomes are present in a cell in a human ovary during each of the following stages

- A 46 Prophase of the first meiotic division
B 23 At the end of telophase of the first division
C 23 Prophase of the second meiotic division
D 23 at the end of telophase of the second division
(4 marks)

Question 2

Describe and explain two differences between the processes of meiosis and mitosis. ^{any 2.}

1 Meiosis - cell division for gametes
Mitosis - for growth & repair

2 Meiosis - PMAT twice
Mitosis - PMAT once

Meiosis - non-identical cells
Mitosis - identical cells

Meiosis - 4 cells
Mitosis - 2 cells
(4 marks)

Question 3

Complete the complementary DNA strand below.

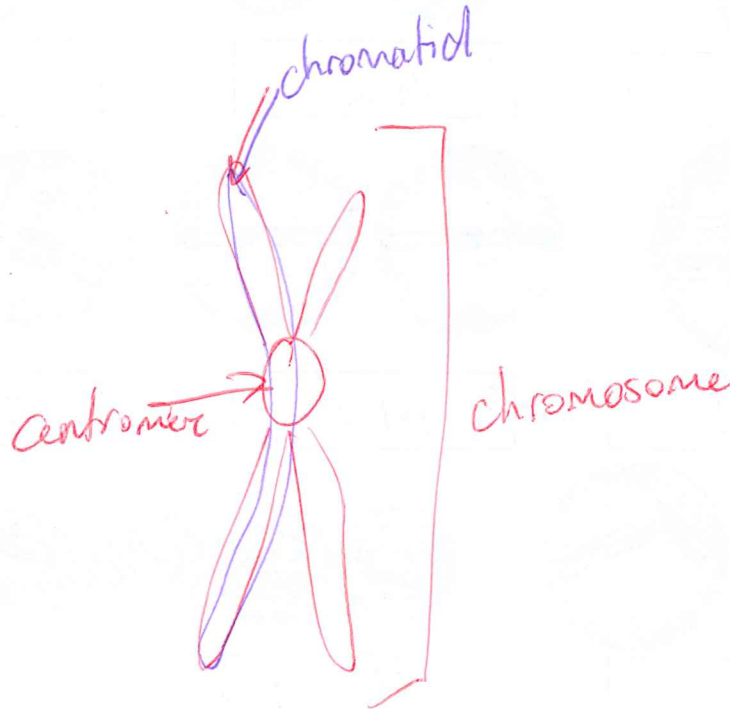
(1 mark)

A T T C G A A C

T A A G C T T G

Question 4

In the space below draw a diagram demonstrating the difference between a chromatid, chromosome and centromere. (3 marks)



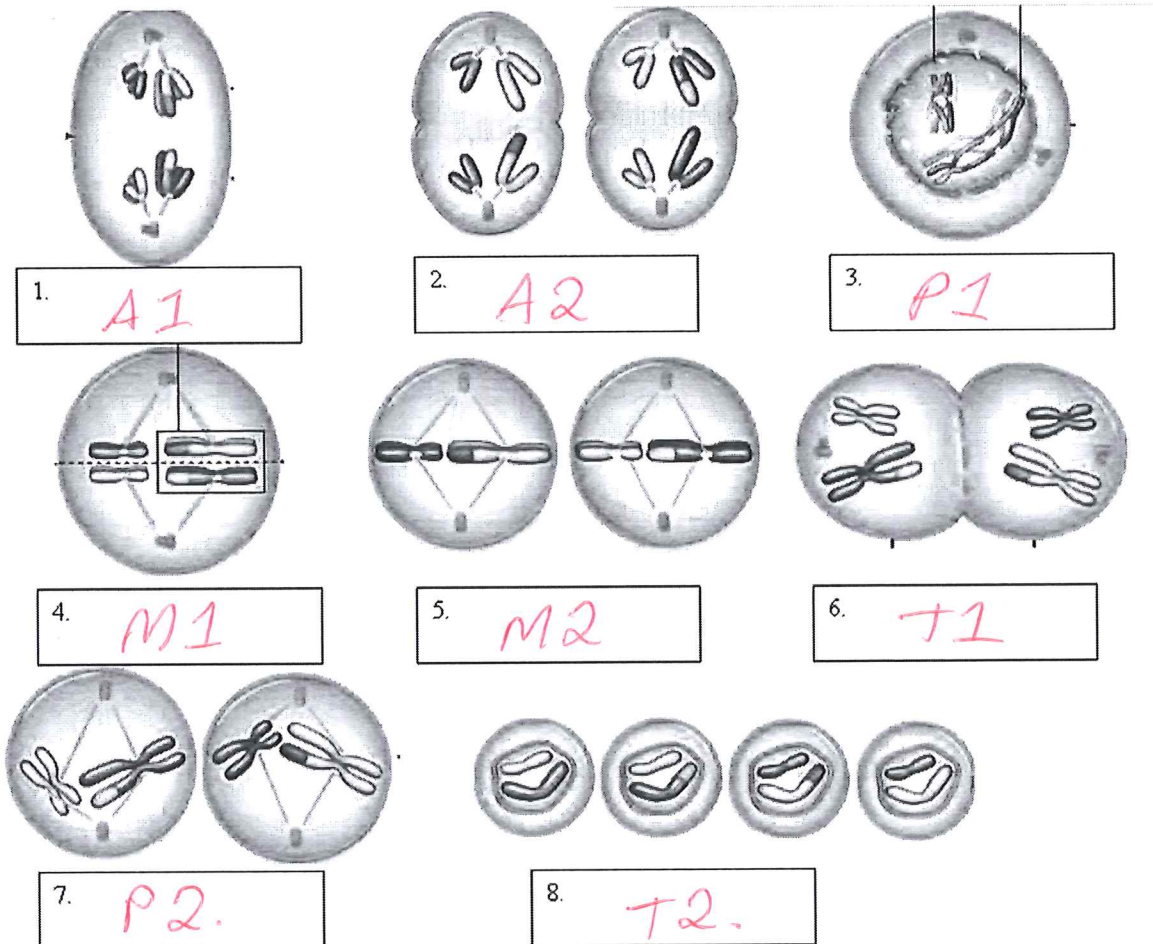
Question 5. Explain the difference between a gene and an allele

(2 marks)

gene = section of DNA that codes
for a protein
allele = alternative form of a gene.

Question 6.

The following are the steps in Meiosis. They are NOT in order. Label each correctly as either Metaphase, Prophase, Telophase or Anaphase followed by the number 1 or 2.
(8 marks)



Question 7. Below is a diagram of DNA. Label 1, 2 and 3.

(3 marks)

